

4-Port TSN ATE/ Compliance Tester Based on NI PXI Platform

Platform Overview

- **Platform:** NI PXI
- **Chassis:** NI PXI-1095
- **Controller:** NI PXIe-8881
- **Traffic Generator/Capture Engine:** NI PXIe-6592
- **Timing & Synchronization:** NI PXIe-6674T
- **Port Count:** 4 x 1G TSN Ethernet ports (line-rate)
- **Standards Supported:** IEEE 802.1AS, 802.1CB, 802.1Qbv, 802.1Qbu, 802.3br, 802.1Qci, 802.1Qav, 802.1Qca

1. Executive Summary

This proposal defines a 4-port, 1G line-rate TSN ATE tester based on an NI PXI modular architecture for automated validation of TSN features across endpoints and switches. The system enables functional compliance and performance characterization of deterministic Ethernet behavior through synchronized traffic generation, precise timing correlation, traffic capture and automated analysis, and reporting.

The solution is optimized for:

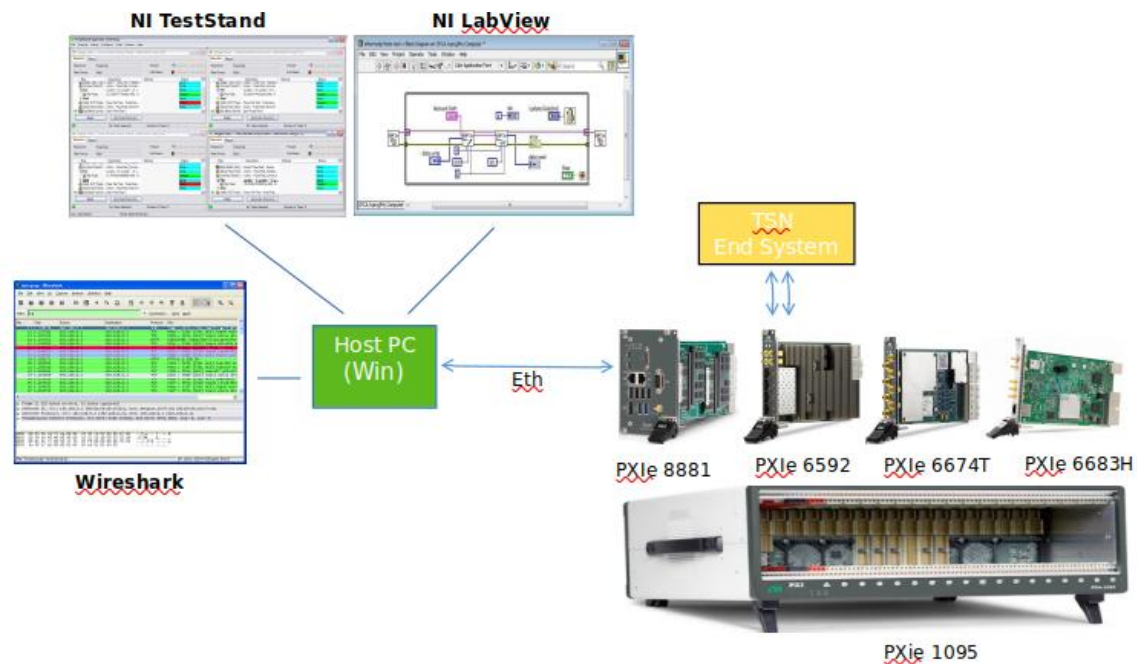
- 1G multi-port concurrent stream testing
- Time-aware shaping (Qbv) schedule verification
- gPTP time synchronization accuracy checks (802.1AS)
- FRER reliability testing (802.1CB)
- Frame preemption (Qbu/802.3br)
- Ingress policing/PSFP (Qci)
- Credit-based shaping (Qav)
- Path control/reservation behavior (Qca)

2. Objectives

- Build a deterministic, PXI-based TSN tester that can generate and analyze 4 simultaneous 1G streams (plus interferers) in real time.
- Validate TSN scheduling, shaping, preemption, policing, redundancy, and time sync compliance.
- Provide repeatable, automated test execution with pass/fail results and engineering debug artifacts.

3. System Architecture

The diagram below shows the complete test setup in the deployment scenario.



System Architecture (4-Port TSN Tester)

a) NI PXI-1095 Chassis

- Hosts controller + timing + traffic/capture hardware
- Allows future slot expansion (more ports, extra capture engines)

b) PXIe-8881 Controller

- Runs NI Linux RealTime for real-time deterministic performance.
- Runs LabView Runtime Engine and TestStand Engine.
- It will take commands over ethernet from TestStand OI and Labview GUI running on a Host PC and execute the tests and send the result back.
- It will host utilities like Tshark to capture the packets in a “.pcap” file and services like “rpcapd” to stream live traffic to host PC over ethernet for analysis.

c) PXIe-6674T Timing Module

- Provides deterministic test start triggers, coherent PXI timebase distribution, phase-aligned stimulus across multiple ports, measurement consistency for long-duration tests

d) PXIe-6592 Traffic Engine

- Used for deterministic and synchronized traffic pattern generation/capture control (with appropriate Ethernet line interface layer).
- Supports building repeatable high-load profiles for TSN features.

e) 4 x 1G Ethernet Line Interface (Port Front-End)

- For true 1G line-rate Ethernet ports, an interface layer is required to connect PXI stimulus/capture to the DUT Ethernet physical ports. This can be implemented via:

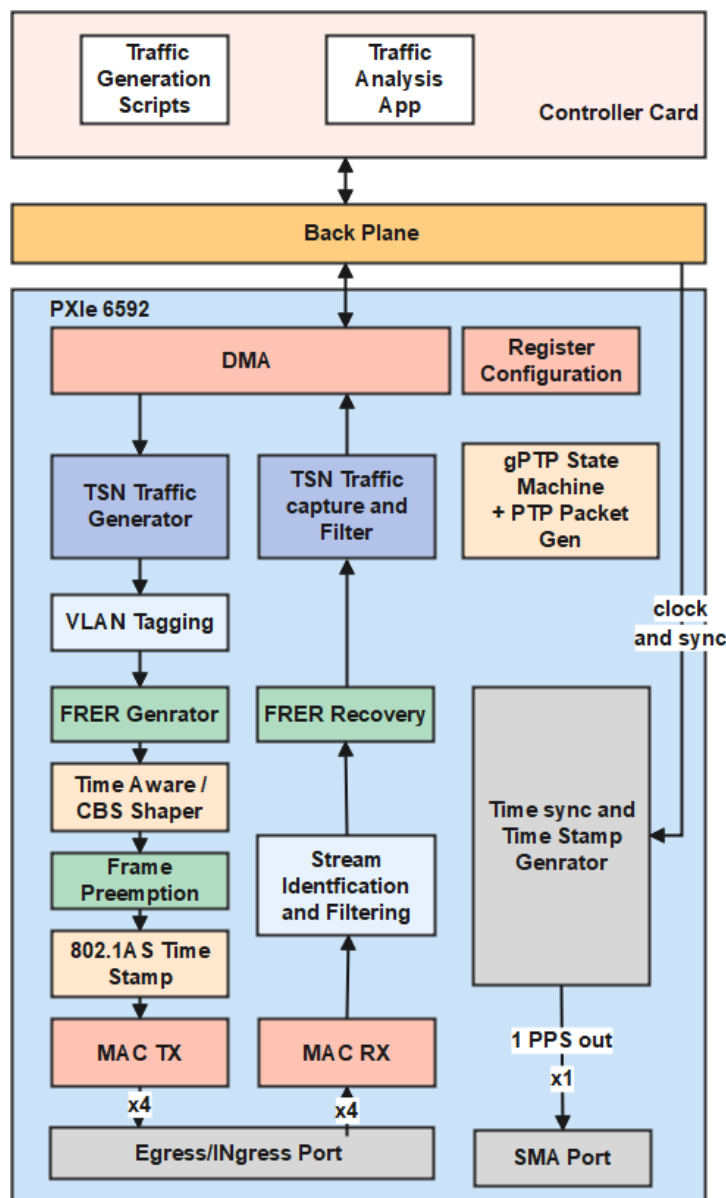
- i) 4-port 1G Ethernet MAC/PHY front-end module(s)
- ii) Copper RJ45 (1000BASE-T) or media variants depending on target DUT.

The tester will be delivered as a 4-port appliance with 1G physical ports presented at the front panel, supporting full-duplex operation.

f) Host PC

- Runs on Windows and interfaces with PXIe-8881 Controller over ethernet.
- Runs Operator Interface (OI) and LabView GUI.
- While TestStand OI will provide the test execution interface for the user, LabView GUI will provide configuration control and status.
- It will additionally run Wireshark GUI to display and analyze captured traffic.

3.2 Architecture Block Diagram



4. Key Capabilities

4.1 Traffic Generation Capabilities

- **Multi-Stream TSN Traffic Profiles**
 - Generate multiple concurrent streams per port with independent VLAN/PCP/DSCP, frame size, rate, burst, and destination patterns
 - Support talker/listener + interferer models (TSN + background BE saturation)
- **Time-Aware and Scheduled Generation (802.1Qbv)**
 - GCL-aligned transmission with configurable base Time, cycle Time, gate masks
 - Ability to generate traffic inside/outside gate windows for negative testing
- **gPTP-Synchronized Traffic (802.1AS)**
 - Generate traffic aligned to gPTP time (time-of-day triggers)
 - Support Sync/Follow_Up/Pdelay traffic emulation (as needed) and time-locked start across ports
- **Frame Preemption Aware Traffic (802.1Qbu / 802.3br)**
 - Generate express vs preemptable classes with controllable interference patterns
 - Create preemption stress conditions (long BE frames + periodic express microbursts)
- **FRER Traffic Generation (802.1CB)**
 - Replicated stream generation across two ports/paths
 - Configurable sequence numbering, wrap/reset behavior, duplicate patterns
 - Ability to intentionally create out-of-order and duplicate bursts
- **PSFP / Policing Stress Traffic (802.1Qci)**
 - Generate streams that violate:
 - CIR/EIR
 - burst limits
 - timing/interval constraints
 - Generate classification challenge traffic (similar MAC/VLAN/PCP but different stream keys)
- **CBS / AVB Traffic (802.1Qav)**
 - Generate AVB/CBS class traffic with controlled rate shaping, bursts, and interference
 - Mixed CBS + BE + TSN scheduled streams for coexistence validation

4.2 Capture & Analysis Capabilities

- Packet loss / mis-order / duplication detection
- Per-stream throughput, latency, and jitter
- Gate schedule alignment verification (Qbv)
- FRER generation and recovery behavior (CB)
- Policing enforcement verification (Qci)
- Preemption effectiveness verification (Qbu/802.3br)
- gPTP time sync monitoring and stability metrics (802.1AS)

4.3 Synchronization & Repeatability

- Synchronized start-of-test across all 4 ports
- Deterministic replay of traffic waveforms and schedules
- Long-run drift and convergence evaluation using the timing module reference

4.4 Error Insertion Capabilities

Frame-Level (Layer 2) Errors:

- Malformed frames (oversized/undersized, invalid lengths)
- Invalid VLAN tags or R-TAGs (for FRER)
- Incorrect checksums or deliberate CRC errors

gPTP Synchronization Errors (IEEE 802.1AS):

- offset injection
- Malformed PTP messages (Sync, Follow_Up, Pdelay_Req/Resp)
- Incorrect timestamps or domain numbers
- Delayed or missing PTP packets
- Path delay asymmetry

Time-Aware Shaper (TAS - IEEE 802.1Qbv) Errors:

- Traffic sent outside scheduled gate windows
- Burst size violations
- Interference during protected slots or guard bands

Frame Preemption Errors (IEEE 802.1Qbu / 802.3br):

- Invalid preemption attempts
- Incomplete or fragmented preemptable frames
- Preemption during non-preemptable periods

FRER Redundancy Errors (IEEE 802.1CB):

- Selective packet loss on one redundant path
- Sequence number gaps, mismatches, or resets
- Duplicate elimination failures

Per-Stream Filtering and Policing (PSFP - IEEE 802.1Qci):

- Flow rate violations (excessive packets)
- Oversized frames exceeding meter limits
- Invalid stream identification

General Network Impairments:

- Variable latency and jitter injection
- Bandwidth throttling or link congestion
- Protocol message corruption (e.g., malformed SRP or MSRP reservations)

5. Compliance Standards Test Coverage

5.1 IEEE 802.1AS (gPTP)

Goal: Validate DUT time synchronization, convergence and stability.

Key test scenarios:

1. gPTP state machines (GM / follower behavior)
2. Best Master Clock Algorithm (BMCA) behavior (if applicable)
3. Sync/Follow-up transmission & reception
4. Peer delay measurement (Pdelay_Req/Resp/Follow_Up)
5. Rate ratio / time recovery stability
6. Clock identity, domain number handling (if used)
7. Timeout behavior, re-sync behavior after link flap
8. Accuracy and stability (offset/jitter under load)
9. Multiple domain operation and domain-specific clock handling
10. Hot-standby / redundant Grand Master (GM) selection and failover behavior

11. Recovery from Grand Master failure (GM change announcement, rapid convergence)
12. Sub-microsecond synchronization accuracy under path asymmetry and load

Measurements: Offset from GM, convergence time, sync stability under load etc.

5.2 IEEE 802.1CB (FRER)

Goal: Validate replication at talker and elimination at listener.

Key test scenarios

1. Replication behavior at talker
2. Elimination and duplicate removal at listener
3. Sequence numbering correctness
4. Recovery from path failure (no application disruption)
5. Latency impact and duplicate detection window behavior
6. Interworking with policing and shaping
7. Behavior with non-disjoint redundant paths
8. Variable delay injection on one redundant path and recovery algorithm performance
9. Sequence history window sizing impact on burst handling and latency

Measurements: duplicate elimination, disorder, recovery after failure

5.3 IEEE 802.1Qbv (Time-Aware Shaper)

Goal: Validate scheduled transmission windows

Key test scenarios:

1. Gate Control List (GCL) execution correctness
2. Cycle time handling and guard bands
3. Queue mapping and priority correctness
4. Traffic class isolation during closed gates
5. Behavior on schedule reconfiguration (dynamic update)
6. Interaction with CBS/Best Effort queues
7. Latency/jitter results per scheduled window
8. Guard band violation detection and prevention of best-effort interference
9. Dynamic Gate Control List (GCL) updates under active traffic load
10. Multi-cycle GCL configurations and long-term schedule stability

Measurements: schedule window conformance, worst-case latency/jitter

5.4 IEEE 802.1Qbu + IEEE 802.3br (Frame Preemption)

Goal: Validate express vs preemptable traffic.

Key test scenarios:

1. Correct Express vs Preemptable traffic separation
2. Fragmentation/reassembly correctness
3. Verify max interference time reductions
4. Interoperability between two devices
5. Correct behavior under errors/CRC/fragment loss
6. Verify link partner negotiation (MAC Merge sublayer)
7. Hold/release mechanism behavior during preemption events

8. Multi-hop reassembly failures and error recovery
9. Verification of maximum interference time reduction

Measurements: express latency improvement, fragment correctness, reassembly failures

5.5 IEEE 802.1Qci (PSFP – Policing)

Goal: Validate ingress filtering and policing per stream.

Key test scenarios:

1. Stream identification rules (which frames belong to which stream)
2. Filtering (drop/pass) based on stream rules
3. Policing actions (rate limiting, max SDU, etc.)
4. Time-based gating (if configured)
5. Counter increments and reporting
6. Correct discard reasons (oversize, wrong timing, excess rate)
7. Time-based gate interaction with stream filters (early/late frame handling)
8. Counter accuracy and overflow behavior under sustained high-rate bursts
9. Policing effectiveness for maximum SDU size and rate exceeding scenarios

Measurements: drop rate vs offered load, isolation correctness

5.6 IEEE 802.1Qav (Credit-Based Shaper)

Goal: Validate reserved bandwidth behavior.

Key test scenarios:

1. IdleSlope / SendSlope behavior
2. Credit accumulation and depletion rules
3. Correct shaping under congestion
4. Bounded latency behavior for shaped class
5. Fairness and no starvation behavior (for other classes)
6. Correct behavior after idle periods

Measurements: Throughput reservation, latency bound, shaping consistency

5.7 IEEE 802.1Qca (Path Control & Reservation)

Goal: Validate explicit path behavior and recovery. Many industrial TSN systems use centralized configuration instead of SRP.

Key test scenarios:

1. Talker advertise / listener ready state correctness
2. Bridge registration and propagation correctness
3. Bandwidth calculation and admission control
4. Correct teardown behavior
5. Compatibility with shaping (especially CBS)
6. Multiple streams and failure scenarios

Measurements: convergence time, flow-to-path mapping, resource exhaustion handling

6. Compliance Test Methodology

Time-Sensitive Networking (TSN) compliance testing ensures that Devices Under Test (DUTs) conform to the relevant IEEE 802.1 standards, delivering deterministic performance in terms of time synchronization, bounded latency, traffic scheduling, redundancy, and resource reservation. The testing methodology must be adaptable to the specific type of DUT and its port configuration, as these factors significantly influence test setup complexity, coverage requirements, and execution strategy. There are two type of TSN devices

- **TSN Endpoints (End Stations):** Typically single-port or dual-port devices (e.g., sensors, actuators, controllers, or embedded TSN endpoints).
 - **Focus Areas:**
 - gPTP follower behavior and peer delay measurement.
 - Scheduled transmission as talker (Qbv Gate Control Lists, CBS credit shaping).
 - Frame replication as talker or elimination as listener (802.1CB).
 - Preemption support (express or preemptable MAC).
 - Ingress policing and stream filtering (802.1Qci).
 - **Test Setup:**
 - Simple point-to-point or star topology. The Tester ports acts as Talker or listener port for DUT.
 - DUT connected to one or two tester ports. Two ports is must for FRER testing if supported by DUT.
 - Minimal port-to-port forwarding considerations.
- **Multi-Port TSN Switches (Bridges):** Variable ports ranging from 4 to 48+ ports (e.g., edge switches, aggregation switches, or backbone switches).
 - **Focus Areas:**
 - gPTP transparent/boundary clock operation with accurate residence time correction across ports.
 - Independent per-port Time-Aware Shaping (Qbv) with complex Gate Control Lists and guard bands.
 - Cut-through and store-and-forward behavior under varying load conditions.
 - Frame Replication and Elimination for Redundancy (FRER) across multiple ingress paths.
 - Per-port stream filtering, policing, and credit-based shaping.
 - Port isolation, VLAN handling etc.
 - **Test Setup:**
 - Complex topologies (mesh) require multiple tester ports.
 - Selective or full mesh connectivity depending on port count.
 - Stress testing with simultaneous traffic on multiple ports.
 - To test switches with more than 4 ports multiple PXIe-6592 cards would be required in the chasis.

Some Example configurations for a 4 port TSN switch as DUT is discussed below.

1) gPTP Convergence

Objective: Validate gPTP convergence time and stability after DUT boot.

Topology (4-port):

- P1 ↔ DUT Port1 (GM reference emulator/time source)

- P2 ↔ DUT Port2 (slave observation)
- P3 ↔ DUT Port3 (background load)
- P4 ↔ DUT Port4 (optional observation)

Procedure:

- Start capture on P2 (and P4 if used).
- Power cycle DUT.
- Start gPTP message generation/monitor from ATE.
- Measure time until steady Sync interval observed at P2.
- Apply background traffic on P3 at 50% load.
- Continue monitoring offset stability for 10 minutes.

Expected Results: Sync/Follow_Up and Pdelay messages conform to requirements; offset stabilizes within acceptable bound; no continuous role flapping.

2) Gate Schedule Enforcement

Objective: Validate TAS gate open/close behavior precisely controls transmissions.

Topology:

- P1 ↔ DUT Port1 scheduled TSN stream.
- P2 ↔ DUT Port2 BE interferer.
- P3 ↔ DUT Port3 captures output.
- P4 ↔ DUT Port4 optional monitor.

Procedure:

- Configure DUT TAS schedule (GCL) with defined TSN and BE windows.
- Start TSN stream on P1 (high PCP).
- Start BE interference on P2 at 90–99% load (low PCP).
- Capture egress on P3 and compute timestamp alignment.
- Verify TSN frames occur only during gate-open window

Expected Results:

- TSN schedule conformance.
- no BE traffic leakage into protected slots
- deterministic latency maintained.

3) Express Traffic Latency Under Congestion

Objective: Confirm express frames preempt preemptable frames and meet latency bounds.

Topology:

- P1 ↔ DUT Port1 transmit preemptable saturation stream
- P2 ↔ DUT Port2 transmits express periodic stream
- P3 ↔ DUT Port3 captures preempted traffic for analysis.
- P4 ↔ DUT Port4 transmits express periodic stream with different traffic class.

Procedure:

- Enable frame preemption on DUT ports/links.
- Start preemptable stream from P3 near 1G using large frames (1518B).
- Start express stream from P1 periodic small frames (128B).
- Capture egress on P2, compute express latency distribution.
- Repeat with preemption disabled and compare latency results.

Expected Results:

- Express latency significantly reduced vs baseline
- no express loss under congestion

- correct fragment/reassembly behavior.

4) FRER: Frame Replication Behavior at Talker

Objective: Verify FRER replication sends equivalent frames on two disjoint paths.

Topology:

- P1 injects stream into DUT
- P2 and P3 capture replicated egress paths
- P4 optional end listener

Procedure:

- Configure DUT to enable FRER replication for Stream S1.
- Start traffic on P1 with sequence numbering enabled.
- Capture traffic on P2 and P3 simultaneously.
- Verify payload match, sequence correctness, and 2 replicas per input frame.

Expected Results:

- Two matching replicas per frame
- no sequence discontinuity beyond expected behavior.

7. Proposed Test Modes

7.1. Compliance Mode

- Strict pass/fail per requirement
- Standard-aligned test steps
- Audit-ready reports

7.2. Automated Regression Mode (CI)

- CLI-driven execution
- Configuration-based suite selection
- JSON/CSV export for dashboards

8. Reports and Outputs

- Test summary with pass/fail per feature
- Detailed KPIs: latency min/avg/max, jitter (PDV), throughput per stream, loss/duplicate/misorder counts
- Evidence artifacts: PCAP (filtered), CSV time-series exports, logs of schedules/policing/sequence tracking